Assignment 1

Write the formula for euclidean distance between points x and y in n-dimension:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Log likelihood function

$$\ell(\theta) = \sum_{i=1}^{n} \log p\left(x_i \mid \theta\right)$$

Probability density function

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} e^{-\frac{t^2}{2\sigma^2}}\, dt$$